

Non-Linear Dynamics Notes

Vyas Mokal

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Lecture 1

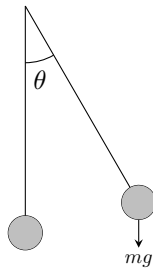
1.1 Books

- Steven H. Strogatz, *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering*, 2nd ed., Westview Press, 2015.

This is Bible for the subject. It is standard, detailed and a bit old text.

1.2 Dynamics

- Study of evolution of systems
Eg. Simple Pendulum



has equation of motion

$$\ddot{\theta} - \frac{g}{l} \sin \theta = 0 \quad (1.1)$$

For small angles

$$\ddot{\theta} - \frac{g}{l} \theta = 0 \quad (1.2)$$

Which is linear. But if θ is not small then non-linear and hence included in NLD¹.

- As equation becomes Non-Linear, the solution drastically changes

¹**NLD:** Non-Linear Dynamics

- Non-Negligible Non-Linear terms are not perturbative and needs separate study
- The subject NLD has advanced after the computers became powerful enough to perform high level computation

1.3 1D Maps

- Suppose we have a sequence of numbers (real numbers mostly)

$$x_0, x_1, x_2, \dots$$

Such that

$$x_0, x_1 = f(x_0), x_2 = f(x_1), \dots \quad (1.3)$$

i.e.

$$x_{n+1} = f(x_n) \quad (1.4)$$

Then it is called an *Map*²

1.3.1 Fixed Point

- Solution of (1.3) such that

$$x^* = f(x^*) \quad (1.5)$$

then x^* is called *fixed point*. Note that it will give rise to the sequence of form

$$x_0, x_1, x_2, \dots, x^*, x^*, x^*, \dots$$

²Note here that the function f remains same